Environment International 208 (2026) 110131



Contents lists available at ScienceDirect

Environment International

journal homepage: www.elsevier.com/locate/envint



Full length article



Prenatal exposure to wildfire-related $PM_{2.5}$ and autism spectrum disorder in children born in California between 2001–2019

Karl O'Sharkey [a], Sanjali Mitra [a], Ting Chow [a], Amanda J. Goodrich [b], Kathryn C. Conlon [b], Miriam Nuño [b], Deborah H. Bennett [b], Sean Raffuse [d], Mohammad Astaneh [e], Yusheng Zhao [e], Michael Kleeman [f], Beate Ritz [a], Rebecca J. Schmidt [b,c,*]

[a] Department of Epidemiology, Fielding School of Public Health, University of California, Los Angeles, CA 90095, USA
[b] Department of Public Health Sciences, School of Medicine, University of California, Davis, CA 95616, USA
[c] MIND Institute, School of Medicine, University of California, Davis, Sacramento, CA 95616, USA
[d] Air Quality Research Center, University of California, Davis, CA 95616, USA
[e] Department of Land, Air, and Water Resources, College of Agriculture and Environmental Science, California, Davis, CA 95616, USA
[f] Department of Civil and Environmental Engineering, College of Engineering, University of California, Davis, CA 95616, USA

ARTICLE INFO

Keywords:
Air pollution
Wildfires
$PM_{2.5}$
Autism Spectrum Disorder
ASD
WUI

ABSTRACT

Background: Wildfire smoke is an increasingly prevalent source of air pollution and contains a complex mixture of neurotoxic constituents that could affect neurodevelopment, yet its potential role in autism spectrum disorder (ASD) etiology remains unexplored. We examined associations between wildfire-specific $PM_{2.5}$ exposures and ASD risk in California births, considering exposure timing, intensity, and background air pollution.

Methods: We analyzed 8.6 million births in California from 2001 to 2019, linking records to ASD diagnoses in California Department of Developmental Services records through 2022. Wildfire-related $PM_{2.5}$ exposures, including biomass burning and wildland-urban-interface (WUI) source tracers, were estimated for preconception, pregnancy, and post-pregnancy periods. Exposure metrics included continuous, percentile-based, and episode-based indicators. Logistic regression was used to estimate odds ratios, adjusting for sociodemographic covariates. Stratified analyses were conducted by background $PM_{2.5}$, urbanicity, diagnostic era, and sex.

Results: Continuous wildfire-specific exposures during the pregnancy-period were weakly associated with ASD, but stronger effects emerged at high exposure percentiles and during intense wildfire episodes ($\geq$ 35 $\mu g/m^3$). Associations were most pronounced among children in the lowest quartile of prenatal background $PM_{2.5}$, with odds of ASD increasing by 50% for those in the $>$ 90th percentile of wildfire $PM_{2.5}$ during pregnancy. WUI-related $PM_{2.5}$ was associated with ASD in metro areas, suggesting unique toxicity. Effects were stronger in non-metro regions, with the exception of WUI-related $PM_{2.5}$, for births occurring after the implementation of DSM-5, and in White children. Sex differences were minimal, but showed stronger effect estimates for WUI-related $PM_{2.5}$ and high exposure episodes for males.

Conclusion: Wildfire smoke exposure during pregnancy may increase ASD risk, particularly in low background air pollution and rural settings. High-intensity wildfire events were most strongly associated with ASD, underscoring the need for targeted policies, such as vegetation management, land use planning, and indoor filtration upgrades in the context of increasing wildfire frequency and severity.

1. Background

Autism spectrum disorder (ASD) is a developmental condition characterized by challenges with communication, restricted or repetitive behaviors, and sensory sensitivities (American Psychiatric Association, 2020). In 2022, the Centers for Disease Control and Prevention (CDC) estimated that 3.2% of children in the United States (US) were diagnosed with ASD by age 8, and the prevalence of ASD varies regionally, ranging from 2.3% in Maryland to 4.5% in California (Shaw et al., 2025). In California, the cumulative incidence of ASD by both ages 4 and 8 has steadily increased over the past three decades. This rise has been shaped by regional and sociodemographic factors, including race,

* Corresponding author.
E-mail address: rjschmidt@health.ucdavis.edu (R.J. Schmidt).

https://doi.org/10.1016/j.envint.2026.110131
Received 6 October 2025; Received in revised form 3 February 2026; Accepted 3 February 2026
Available online 6 February 2026
0160-4120/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

ethnicity, and area-level socioeconomic status (O'Sharkey, Mitra, et al., 2024). Such variation has prompted researchers to identify risk factors that may differ across population groups, with air pollution (AP) increasingly recognized as an important environmental contributor. This is particularly relevant given that ASD typically manifests early in life, suggesting that *in-utero* or early-life exposures to AP may be especially important.

Regulated by the Environmental Protection Agency (EPA) under the Clean Air Act, criteria air pollutants, a set of six common pollutants for which the EPA sets national air quality standards, are among the most commonly studied in relation to health outcomes. Among these, fine particulate matter < 2.5 μm ($PM_{2.5}$) and nitrogen dioxide (NO_2) have been most frequently investigated for their potential role in ASD (Becerra et al., 2013; Rahman et al., 2022; Ritz et al., 2018). Meta-analyses suggest a small positive association between exposure to these pollutants and ASD risk, though estimates from individual studies vary substantially (Chun et al., 2020; Dutheil et al., 2021). This heterogeneity may reflect differences in study design, exposure assessment, variable confounding adjustment, or a failure to account for the specific composition and sources of pollutants. For instance, NO_2 often co-occurs with other traffic-related emissions, and $PM_{2.5}$ is a complex mixture whose health impacts likely depend on its chemical makeup (Beckerman et al., 2008; Oroumiyeh et al., 2022). Our previous research found that traffic-related $PM_{2.5}$ components and oxidative stress potential were associated with ASD in Los Angeles (O'Sharkey, Mitra, et al., 2024), underscoring the need to evaluate source-specific exposures.

While most ASD-related air pollution studies have focused on urban or traffic sources, wildfire smoke has emerged as an increasingly important, and largely underexamined, source of $PM_{2.5}$ with distinct exposure characteristics and biological relevance. For example, wildfire emissions differ in composition and toxicity (Wegesser et al., 2009), often containing elevated levels of organic carbon, volatile organic compounds, and polycyclic aromatic hydrocarbons, and are difficult to avoid during large fire events (Schlosser et al., 2017; Urbanski et al., 2008). As wildfire frequency and intensity rise (Cunningham et al., 2024; Williams et al., 2019), wildfire smoke represents an increasingly prominent source of population-level air pollution. Recent analyses of the January 2025 Los Angeles wildfires demonstrate that wildfire smoke events can produce sustained, region-wide elevations in particulate air pollution across dense urban areas, with exposure levels and durations exceeding those typically examined in ambient air pollution studies (Schollaert et al., 2025). These exposure characteristics underscore the need to understand potential impacts during sensitive development periods, including those relevant to neurodevelopment.

The mechanisms through which wildfire smoke exposure could contribute to ASD risk are likely to be complex and multifactorial. Evidence from studies of prenatal ambient $PM_{2.5}$ exposure indicates that neurodevelopmental effects vary by gestational timing and sex, highlighting the importance of exposure timing during sensitive developmental windows (Chiu et al., 2016). Although wildfire-specific neurodevelopmental studies are limited, reviews of wildfire smoke and related combustion exposures suggest that inflammation, oxidative stress, endocrine disruption, and placental dysfunction represent biologically plausible pathways through which wildfire smoke could influence fetal brain development (Basilio et al., 2022). Though wildfire smoke is known to affect pregnancy (Basilio et al., 2022), respiratory (Wilgus & Merchant, 2024), and cardiovascular health (Heaney et al., 2022) in humans, its potential role in ASD etiology has received little attention to date, despite plausible mechanisms, such as oxidative stress and inflammation (Fang et al., 2023; White, 2024). Experimental studies in rhesus macaques provide strong biological support, linking early gestational wildfire smoke exposure to increased pregnancy loss, systemic inflammation, altered stress hormone regulation, and persistent behavioral and memory impairments in offspring, as well as long-term epigenetic modifications in genes involved in immune function and neurodevelopment, including synaptogenesis (Brown et al., 2022; Capitanio et al., 2022; Willson et al., 2021). The potential effects of exposure during critical developmental periods, including preconception, individual trimesters of pregnancy, and the post-pregnancy years, also remain poorly understood.

In this study, we examine the impact of air pollution exposure during the preconception, pregnancy, and post-pregnancy periods on ASD risk in children born in California between 2001 and 2019, with a particular focus on different types of wildfire smoke, including particulate matter generated by combustion of regular biomass, and combustion of manufactured materials at the wildland-urban interface (WUI). We also assess how sociodemographic factors, including race/ethnicity and socioeconomic status, and regional factors, such as urbanicity, may modify the relationship between wildfire exposure and ASD diagnosis, aiming to identify vulnerable populations and inform targeted policy and intervention strategies. To our knowledge, this is the first large-scale, population-based study to evaluate the association between wildfire smoke exposure and ASD, addressing a critical gap in the environmental neurodevelopmental literature.

2. Methods

2.1. Study population and ASD identification

We obtained birth records from the California Department of Public Health (CDPH) for all live births occurring between January 1, 2001, and December 31, 2019. Eligible children were born to mothers residing in California, ensuring alignment with exposure assessment boundaries. From the initial cohort of 9,735,275 children born in the assessment region, we excluded records with missing (N = 259,170) or implausible gestational age values (< 140 or > 315 days; N = 109,557), missing or implausible birth weights outside the range of 500 to 6,800 g (N = 9,006 (missing = 57)), and non-singleton births (N = 288,481). We further restricted the sample to records with complete data on key covariates (N = 440,692). The final analytical dataset included 8,628,369 children.

ASD diagnoses were identified through the California Department of Developmental Services (DDS), which operates a network of 21 regional centers covering the entire state. DDS data were available from January 1, 2001, through December 31, 2022, allowing for a minimum follow-up period of three years for all children in the cohort. Referrals to DDS typically follow developmental screenings or assessments conducted by pediatricians, psychologists, educators, or other qualified specialists. Prior research has estimated that approximately 75–80% of children with ASD in California are enrolled in DDS (Croen et al., 2002).

We linked DDS records to birth certificates using probabilistic record linkage via the CDC's Link Plus software (Link Plus | Registry Plus | CDC, 2023). Matching variables included child's and mother's names, birthdates, sex, and zip code. Record pairs were evaluated based on match score and consistency of identifying variables, including name, date of birth, and sex. Duplicate matches were resolved through additional verification using parent and child identifiers. In total, 84.4% of DDS records were successfully linked to California birth records. Unlinked cases were primarily due to missing California birth certificates, likely the result of being born out of state, or incomplete identifying information in DDS records.

2.2. Air pollution exposures

Air pollution exposures were estimated using a hybrid modeling approach that captured both total ambient pollution and wildfire-specific contributions. At a daily temporal resolution, total pollutant concentrations were derived by combining outputs from a source-resolved chemical transport model (CTM) with ground-level observations through Random Forest Regression (RFR). The CTM, developed by the University of California, Davis/California Institute of Technology (UCD/CIT), simulated the atmospheric transport and chemical transformation of wildfire emissions, distinguishing smoke contributions

from both rural and urban fires (Hu et al., 2014, 2017; Venecek et al., 2019; Yu et al., 2019). Emissions inputs were derived from satellite-based burned area detections using the BlueSky modeling framework (Yao et al., 2013), with a novel addition for this project being the incorporation of daily-resolved wildfire activity based on reported burn perimeters. Although the UCD/CIT CTM tracks particle dynamics across 15 size bins (0.01–10 μm), our study focused specifically on the $PM_{2.5}$ size fraction. Model outputs were generated on a 4 km × 4 km grid across the majority of California, with a coarser 24 km × 24 km grid only in the most remote, sparsely populated regions where fewer births occur. To isolate wildfire contributions, source apportionment techniques were applied to differentiate smoke from other emission sources such as vehicular traffic and cooking (Yu et al., 2019). Particles emitted from biomass burning differ substantially in both size distribution and chemical composition from particles emitted by anthropogenic sources such as gasoline, diesel, aircraft, and food cooking, providing a physical and chemical basis for distinguishing wildfire-related $PM_{2.5}$ from typical urban pollution (Kleeman et al., 2008). Secondary organic aerosol formation was modeled by incorporating gas-phase reactions that produce semi-volatile compounds, which subsequently partition to the particle phase based on established atmospheric chemistry mechanisms (Cappa et al., 2016; Jathar et al., 2015). The SAPRC11 chemical mechanism was used to simulate oxidant levels that drive secondary particle formation (Carter & Heo, 2013). Particle dynamics such as Brownian coagulation and size-dependent deposition velocities, driven by both diffusivity and gravitational settling, were also included (Seinfeld & Pandis, n.d.). The resulting UCD/CIT exposure fields have been used previously in epidemiological analyses of outcomes such as mortality, low birth weight, and ASD (Laurent et al., 2016; Ostro et al., 2015; Yang et al., 2024). For this study, model predictions were corrected for residual bias using RFR informed by satellite-based aerosol optical depth (AOD), meteorological data, source-specific emission tracers, and $PM_{2.5}$ measurements from low-cost sensors. Mean fractional bias during wildfire exposure days was maintained below 3%.

2.2.1. Wildfire metrics

Wildfire smoke exposure was characterized using both concentration-based and event-based metrics. We examined total wildfire-sourced $PM_{2.5}$, estimated by summing two established chemical source tracers (biomass burning and WUI), during multiple developmental windows (preconception, each trimester, entire-pregnancy, and post-pregnancy year). Biomass burning particles have a distinct chemical profile relative to urban anthropogenic $PM_{2.5}$. They are dominated by organic carbon, polycyclic aromatic compounds, and biomass-specific tracers such as levoglucosan, potassium, and rubidium, with particle mass typically peaking around 300 nm (Kleeman et al., 2008). The biomass burning tracer primarily reflects particulate emissions from combustion of vegetative fuels, including grasses, shrubs, and forest biomass, which represent the dominant fuels in most California wildfires, although the tracer may also capture a smaller contribution from residential wood burning during winter months. In addition to modeling these concentrations as continuous variables (scaled by interquartile range (IQR)), we also created percentile-based categorical variables to capture relative exposure levels. For the entire-pregnancy period, each wildfire (WF)-specific pollutant (WF-$PM_{2.5}$, WF-Biomass Burning, WF-WUI) was categorized into four exposure groups based on its concentration distribution: ≤ 30th percentile (reference), 30–60th percentile, 61–90th percentile, and > 90th percentile. We used these percentile cut points to ensure that the highest exposure category captured the extreme wildfire-related $PM_{2.5}$ concentrations while preserving sufficient case counts within each group. Next, for entire-pregnancy and 1-year post-pregnancy, we created several wildfire smoke episode metrics based on two concentration levels, 9 $\mu g/m^3$ or 35 $\mu g/m^3$ thresholds that correspond to the National Ambient Air Quality Standards for $PM_{2.5}$ annual average and 24-hour average, respectively (CARB, n.d.). For each child, we computed:

1) Wildfire day counts: The total number of days during each exposure window when wildfire $PM_{2.5}$ exceeded 9 $\mu g/m^3$ or 35 $\mu g/m^3$.
2) Episode indicators: The number of days on which the pregnancy/child experienced wildfire smoke episodes, defined as any day with wildfire $PM_{2.5} \geq 9$ $\mu g/m^3$ or ≥ 35 $\mu g/m^3$ that was proceeded by a 3-day or 7-day washout period (defined as days below the threshold). We used 3-day and 7-day washout periods to identify distinct wildfire smoke episodes, requiring several consecutive days below the threshold before counting a new event; these intervals were selected to capture both short and slightly longer gaps between unique wildfire-related exposure events.

2.3. Statistical analysis

We estimated odds ratios (ORs) and 95% confidence intervals (CIs) for the association between $PM_{2.5}$ mass and wildfire-related air pollution exposures and ASD using logistic regression models. For continuous exposure metrics, we assessed associations across multiple windows, including preconception (three months before pregnancy), each trimester of pregnancy, the entire-pregnancy period, and 1-year post-pregnancy period (up to one year after birth). For categorical exposure metrics, we focused on entire-pregnancy and post-pregnancy exposures only. To inform ongoing discussions around mutual adjustment for temporally correlated exposure periods, we also fit models adjusting simultaneously for entire-pregnancy and post-pregnancy exposure. The correlation between entire-pregnancy and post-pregnancy wildfire $PM_{2.5}$ exposures was moderate ($r = 0.51$), suggesting these windows capture distinct temporal variation, which is expected given the highly episodic nature of wildfire smoke. As results were comparable, all other analyses used single-window models.

To assess source-specific associations independent of background air pollution, we ran models both with and without adjustment for non-WF-$PM_{2.5}$ mass. All continuous wildfire-related pollutants were scaled to an IQR of the exposure distribution across the study population. A Pearson correlation matrix was calculated among all $PM_{2.5}$ mass and WF-related pollutant exposure variables to assess potential collinearity.

Covariates were selected based on prior literature and included sex (male, female), parity (nulliparous vs. parous), maternal age (< 20, 20–24, 25–29, 30–35, > 35 years), race/ethnicity (Hispanic, Non-Hispanic White, Non-Hispanic Black, Asian or Pacific Islander, American Indian or Alaska Native, Other), maternal place of birth (U.S.-born vs. foreign-born), maternal education level (≤ 8th grade, some high school, high school diploma or GED, some college, associate's degree, bachelor's degree, graduate degree), type of prenatal insurance (none, government, private, self-pay, other), area-level socioeconomic status (assigned at the census block group), and indicator variables for each birth year (2001–2019).

We conducted stratified analyses to explore whether associations between wildfire smoke exposure and ASD varied across subgroups defined by background air pollution levels, urbanicity, changes in diagnostic practices, infant sex, race/ethnicity, and area-level socioeconomic status to identify populations that may be more or less affected by wildfire-related air pollution. Specifically, we stratified by: (1) background $PM_{2.5}$, defined as $PM_{2.5}$ mass composed of primary $PM_{2.5}$ source tracers excluding primary WF-related tracers; (2) urbanicity, using Rural-Urban Commuting Area (RUCA) codes categorized as metropolitan (RUCA 1–3) or non-metropolitan (RUCA 4–10) (Dobois & Sanders, n.d.); (3) the introduction of DSM-5 criteria into the California Department of Developmental Services (DDS) system around November 2014, with children born between 2014 and 2019 classified as the post-DSM-5 cohort, as they would likely have been evaluated under the new criteria given that children under one year of age are rarely assessed for ASD (Nevison et al., 2018); (4) infant sex; (5) race/ethnicity; and (6) area-level socioeconomic status (Yang et al., 2020). Separate logistic regression models were estimated within each stratum and adjusted for the same covariates as the main analyses to assess consistency and

potential modification of associations. To reduce spurious detection of statistically significant interactions driven by large sample size and narrow confidence intervals, we restricted formal interaction testing to contrasts where stratum-specific odds ratios differed by more than 5%.

This research received approval from the University of California, Davis Institutional Review Board, the California Committee for the Protection of Human Subjects, and the University of California, Los Angeles Office of the Human Research Protection Program. All analyses were performed using R software (version 4.1.2; R Foundation for Statistical Computing, Vienna, Austria).

3. Results

3.1. Study population characteristics

This study included a large population of children born between 2001 and 2019, comprising 118,339 diagnosed with ASD and 8,510,030 without ASD, all followed for diagnoses through 2022 (Table 1). Children diagnosed with ASD were more likely to be male (79.9%) compared to 50.8% in the non-ASD population. Hispanic children made up the largest racial and ethnic group overall. A higher proportion of children diagnosed with ASD were Black (7%) compared to those without ASD (5.6%). Mothers of children with ASD tended to be older at the time of delivery, with 50.2% aged 30 or older, compared to 45.2% among mothers of non-ASD children. First-time motherhood was also more prevalent in the ASD group (44.8% vs. 39.3%). The prevalence of ASD increased over time, with higher diagnosis rates among children born in more recent years.

3.2. Wildfire-Related air pollution exposures

The average $PM_{2.5}$ mass concentration during pregnancy was 11.64 $\mu g/m^3$. For wildfire-specific pollutants, the average entire-pregnancy concentration was 1.60 $\mu g/m^3$ for WF-$PM_{2.5}$, 1.57 $\mu g/m^3$ for the biomass burning component, which constitutes the majority of WF-$PM_{2.5}$, and 0.02 $\mu g/m^3$ for the WUI-related component (**Table S1**). The very low concentrations observed for the WUI tracer (e.g., median of 0.003 $\mu g/m^3$) reflect minimal or no exposure for most of pregnancy, with brief high-intensity spikes likely driving elevated percentiles. These values may represent only a few days of elevated WUI-related smoke averaged over the entire pregnancy period.

In terms of event-based metrics, the mean number of days on which wildfire-sourced $PM_{2.5}$ exceeded 9 $\mu g/m^3$ during entire-pregnancy was 7.31 with a maximum of 112 days, and the number of days exceeding 35 $\mu g/m^3$ averaged 0.4 days with a maximum of 71 days. The number of wildfire smoke episodes (defined by isolated exceedances following 3- or 7-day washout periods) ranged from 0 to 20 events per entire-pregnancy, depending on the threshold and washout definition, in addition to geographic differences. WF-$PM_{2.5}$ was strongly correlated with its biomass burning component ($r = 0.99$), reflecting the fact that biomass burning comprises the majority of WF-$PM_{2.5}$, and moderately correlated with WUI-related concentrations ($r = 0.29$) (**Table S2**).

3.3. Wildfire-Related air pollution exposure and Autism spectrum disorder

Table 2 presents single-time period associations between an IQR increase in wildfire-related exposures and ASD risk based on logistic regression models. Separate models were run for each exposure metric and developmental window. Single-pollutant WF-$PM_{2.5}$ models showed no associations with ASD across any of the windows of exposure. This was in contrast to the effect estimates for $PM_{2.5}$ mass overall, which were elevated across all exposure windows, with the entire-pregnancy estimate having the strongest association (OR: 1.10; 95% CI: 1.09, 1.11). For WF-$PM_{2.5}$, results did not materially change after adjusting for non-wildfire-related $PM_{2.5}$ mass in most co-adjusted models. Models for the biomass burning and WUI source tracers showed similarly null findings across all time points, and only for entire-pregnancy and preconception biomass burning we found small positive associations in the models co-adjusted for non-wildfire-related $PM_{2.5}$ mass (OR: 1.02; 95% CI: 1.02, 1.03, and 1.01 (95% CI: 1.01, 1.02, respectively). Results from models mutually adjusted for entire-pregnancy and post-pregnancy exposures were similar to single-window models, except for $PM_{2.5}$ mass which was only elevated in the pregnancy period (**Table S3**).

Table 1
Characteristics of Study Population by Autism Spectrum Disorder Status (2001–2019).

Demographics		Total Population: N (%)	ASD: N (%)	Non-ASD: N (%)
	Total	8,628,369	118,339	8,510,030
Sex	Male	4,420,448 (51.2)	94,585 (79.9)	4,325,863 (50.8)
	Female	4,207,921 (48.8)	23,754 (20.1)	4,184,167 (49.2)
Race/Ethnicity	White	2,520,341 (29.2)	31,532 (26.7)	2,488,809 (29.3)
	Hispanic	4,375,568 (50.7)	61,175 (51.7)	4,314,393 (50.7)
	Black	485,284 (5.6)	8,332 (7.0)	476,952 (5.6)
	API	1,202,417 (13.9)	16,655 (14.1)	1,185,762 (13.9)
	AIAN/Other	44,759 (0.5)	645 (0.6)	44,114 (0.5)
Maternal Age	< 20	652,522 (7.6)	6,041 (5.1)	646,481 (7.6)
	20–24	1,779,329 (20.6)	21,921 (18.5)	1,757,408 (20.7)
	25–29	2,293,276 (26.6)	30,970 (26.2)	2,262,306 (26.6)
	30–34	2,294,445 (26.6)	32,373 (27.4)	2,262,072 (26.6)
	≥ 35	1,608,797 (18.6)	27,034 (22.8)	1,581,763 (18.6)
Maternal Education	≤ 8th grade	680,257 (7.9)	6,137 (5.2)	674,120 (7.9)
	9th – 12th	1,314,757 (15.2)	15,507 (13.1)	1,299,250 (15.3)
	High School Grad	2,250,920 (26.1)	32,759 (27.7)	2,218,161 (26.1)
	College (any)	3,493,072 (40.4)	53,586 (45.3)	3,439,486 (40.4)
	Grad degree	889,363 (10.3)	10,350 (8.8)	879,013 (10.3)
Nativity	U.S. Born	4,987,506 (57.8)	70,380 (59.5)	4,917,126 (57.8)
	Foreign Born	3,640,863 (42.2)	47,959 (40.5)	3,592,904 (42.2)
Prenatal Care Payment	None	38,270 (0.4)	445 (0.4)	37,825 (0.4)
	Government	4,084,964 (47.3)	58,388 (49.3)	4,026,576 (47.3)
	Private	4,189,937 (48.6)	56,728 (47.9)	4,133,209 (48.6)
	Self-pay	225,957 (2.6)	1,363 (1.2)	224,594 (2.6)
	Other	89,241 (1.0)	1,415 (1.2)	87,826 (1.0)
Birth Order	First child	3,398,404 (39.4)	53,058 (44.8)	3,345,346 (39.3)
	Subsequent child	5,229,965 (60.6)	65,281 (55.2)	5,164,684 (60.7)
Birth Year (DSM-5 Initiation in CA)	2001–2013	6,110,195 (70.8)	71,843 (60.7)	6,038,352 (71.0)
	2014–2019	2,518,174 (29.2)	46,496 (39.3)	2,471,678 (29.0)

Notes: ASD = autism spectrum disorder; API = Asian or Pacific Islander, AIAN = American Indian or Alaska Native.

Table 2
Wildfire-Related $PM_{2.5}$ Exposure and Autism Spectrum Disorder Risk (N = 8,628,369).

		Single Pollutant Model			Adjusted for Non-WF $PM_{2.5}$ Mass		
Pollutant	**Time Point**	**OR**	**LCI**	**UCI**	**OR**	**LCI**	**UCI**
$PM_{2.5}$ Mass	Pre	1.06	1.05	1.07	–	–	–
	Entire-Pregnancy	1.10	1.09	1.11	–	–	–
	T1	1.07	1.06	1.08	–	–	–
	T2	1.06	1.05	1.07	–	–	–
	T3	1.04	1.04	1.05	–	–	–
	1-Year Post	1.08	1.07	1.09	–	–	–
Wildfire (WF) – Related $PM_{2.5}$	Pre	1.00	1.00	1.01	1.01	1.01	1.02
	Entire-Pregnancy	1.00	1.00	1.01	1.02	1.02	1.03
	T1	1.00	1.00	1.01	1.01	1.00	1.01
	T2	1.00	1.00	1.01	1.01	1.00	1.01
	T3	1.00	0.99	1.00	1.00	1.00	1.01
	1-Year Post	0.98	0.97	0.98	1.01	1.00	1.01
WF-Biomass Burning	Pre	1.00	1.00	1.01	1.01	1.01	1.02
	Entire-Pregnancy	1.00	1.00	1.01	1.02	1.02	1.03
	T1	1.00	1.00	1.01	1.01	1.00	1.01
	T2	1.00	1.00	1.01	1.01	1.00	1.01
	T3	1.00	0.99	1.00	1.00	1.00	1.01
	1-Year Post	0.98	0.97	0.98	1.01	1.00	1.01
WF ∩ Wildland Urban Interface	Pre	1.00	1.00	1.00	1.00	1.00	1.00
	Entire-Pregnancy	1.00	1.00	1.00	1.00	1.00	1.00
	T1	1.00	1.00	1.00	1.00	1.00	1.00
	T2	1.00	1.00	1.00	1.00	1.00	1.00
	T3	1.00	1.00	1.00	1.00	1.00	1.00
	1-Year Post	1.00	1.00	1.00	1.00	1.00	1.00

Notes: Adjusted for infant sex, birth order, maternal age, nativity, race/ethnicity, education, payment of prenatal care, area-level socioeconomic status, and birth year. OR for an IQR increase.

Using percentile-based exposure metrics during the entire-pregnancy period, notable associations were observed only for the WUI tracer, particularly at higher percentile categories (Fig. 1, **Table S4**). In contrast, $PM_{2.5}$ wildfire mass and biomass burning components showed null or inverse associations across percentile groups in single pollutant models; however, when co-adjusted for non-wildfire-related $PM_{2.5}$ mass, > 90th percentile exposures were associated with a 6% increase in odds of ASD. For the threshold-based event metrics, which reflect the number of days or episodes exceeding specific wildfire $PM_{2.5}$ concentrations, elevated odds of ASD were observed primarily at the more extreme ≥ 35 $\mu g/m^3$ threshold. Associations for the ≥ 9 $\mu g/m^3$ cutoff, whether based on daily counts or isolated episodes, were generally weak in the inverse direction or null. Results did not change when co-adjusting for non-wildfire-related $PM_{2.5}$ mass.

Associations between wildfire-related exposures and ASD risk varied across levels of background $PM_{2.5}$ mass (Fig. 2, **Table S5**). In Q1 (lowest quartile of background air pollution), during the entire-pregnancy period, WF-$PM_{2.5}$ and biomass burning source tracer were associated with increased odds of ASD (ORs: 1.07 for both), and the highest WF-$PM_{2.5}$ category (> 90th percentile) was associated with a substantially increased risk (OR: 1.50; 95% CI: 1.44, 1.57). WUI-related exposures also showed positive associations in Q1, particularly in the highest exposure category (OR: 1.21; 95% CI: 1.15, 1.27). In Q2, patterns were similar, but effect estimates attenuated. For WF episodes ≥ 35 $\mu g/m^3$ with 3- and 7-day washout periods, associations remained elevated (ORs: 1.14 and 1.14), as did the > 90th percentile category for all WF exposure percentile-based metrics, suggesting that more intense wildfire events are still associated with ASD even under moderate background air pollution. By Q3, associations were weaker or null across most metrics. Entire-pregnancy WF-$PM_{2.5}$, biomass burning, and WUI tracers no longer showed elevated odds. In fact, WF-$PM_{2.5}$ and the biomass burning tracer percentile-based metrics dipped below null. In Q4 (highest quartile of background $PM_{2.5}$), wildfire-related exposures were generally negatively associated with increased ASD risk, with the exception of the WUI tracer effect estimates remaining slightly elevated in the 61st–90th percentile (OR: 1.06; 95% CI: 1.02, 1.10) and 31st–60th percentile groups (OR: 1.05; 95% CI: 1.01, 1.09).

Associations between wildfire-related exposures and ASD risk differed by urbanicity (**Table S6**). In non-metro areas, where background $PM_{2.5}$ levels are lower on average (**Table S11**), nearly all wildfire exposure metrics, both concentration- and event-based, were associated with elevated odds of ASD. For instance, in non-metro areas, WF-$PM_{2.5}$ during the entire-pregnancy period had an OR of 1.05 (95% CI: 1.04, 1.06), with similar findings for the biomass burning component. Associations were strongest at higher percentile categories: for > 90th percentile WF-$PM_{2.5}$ exposure, the OR was 1.71 (95% CI: 1.57, 1.87), and for biomass burning it was 1.69 (95% CI: 1.55, 1.85). In contrast, children living in metro areas showed either null or inverse associations for the same exposures. One exception was the WUI tracer, which was positively associated in all percentile categories compared to the reference group with ASD in metro areas, while for non-metro areas, only the > 90th percentile of the WUI tracer had a positive association with ASD risk. Event-based wildfire exposures followed a similar pattern: estimates were most consistently elevated in non-metro areas, especially for episodes exceeding 35 $\mu g/m^3$ with 3- or 7-day washout periods (OR: 1.13; 95% CI: 1.08, 1.18 for entire-pregnancy 7-day wash). In metro areas, estimates for these same metrics were lower, though still above 1.0 for the ≥ 35 $\mu g/m^3$ threshold episode metrics in the entire-pregnancy exposure window but not in the post-pregnancy period.

Associations between wildfire-related exposures and ASD risk differed before and after the transition to DSM-5 criteria within California DDS (**Table S7**). In the earlier period (2001–2013), most wildfire exposure metrics were not associated or had inverse associations with ASD. For example, continuous measures of WF-$PM_{2.5}$ and biomass burning during the entire-pregnancy window both showed ORs of 0.97–0.98, and associations remained below null across percentile-based categories. Notably, even the highest WF-$PM_{2.5}$ category (> 90th percentile) was associated with a decreased odds of ASD (OR: 0.89; 95% CI: 0.86, 0.91). In contrast, the WUI tracer showed positive associations across all percentile categories, with the strongest increase observed in the 61st-90th percentile group (OR: 1.11; 95% CI: 1.08, 1.13) followed by the > 90th percentile (OR: 1.08; 95% CI: 1.04, 1.12). In the post-DSM-5 initiation period (2014–2019), WF-$PM_{2.5}$ and biomass burning tracer during the entire-pregnancy window were both associated with elevated odds of ASD using the continuous IQR metric (ORs: 1.05 and 1.04, respectively). All percentile groups for both WF-$PM_{2.5}$ and biomass burning showed elevated odds of ASD compared to the ≤ 30th percentile reference group, with the strongest associations observed in the > 90th percentile group (e.g., ORs: 1.20; 95% CI: 1.15, 1.26 and 1.21; 95% CI: 1.16, 1.27, respectively). In contrast, the WUI tracer was only associated with elevated odds in the > 90th percentile group (OR: 1.14; 95% CI: 1.10, 1.18). Event-based exposure metrics also reflected a temporal shift in associations with ASD. In the earlier period (2001–2013), associations for these metrics were modest or null, with slight elevations observed only for higher-intensity events, for example, WF episodes ≥ 35 $\mu g/m^3$ with a 7-day washout during entire-pregnancy (OR: 1.04; 95% CI: 1.03, 1.06). In the post-DSM-5 initiation period (2014–2019), stronger associations were observed only during entire-pregnancy (OR: 1.13; 95% CI: 1.11, 1.16 for the same metric), while estimates for the postnatal year were null or inverse.

Associations between wildfire exposure and ASD were largely similar across sexes, though a few differences emerged (**Table S8**). For the WUI tracer, positive associations were observed for both males and females, but appeared slightly stronger among males, particularly in the

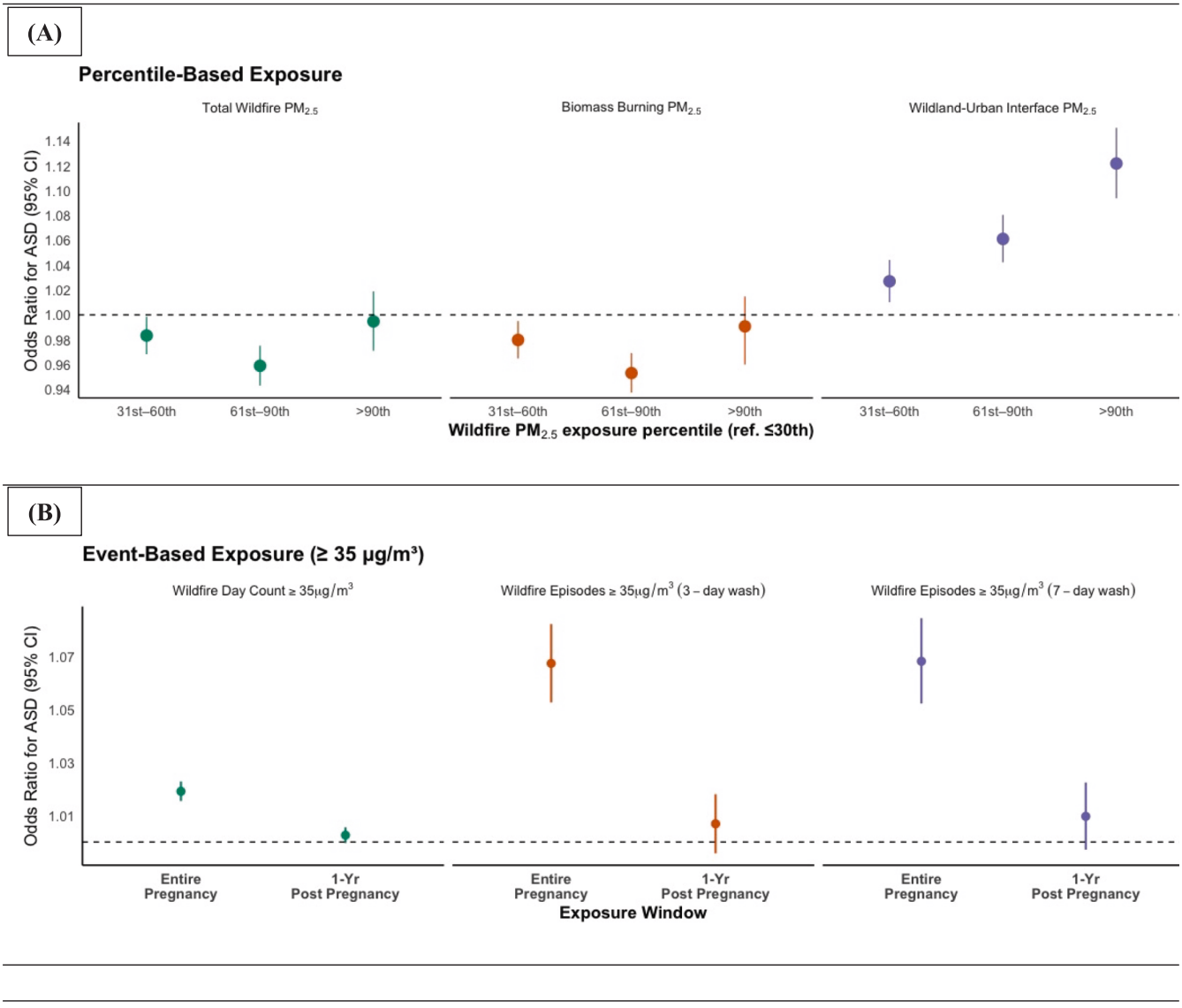


Fig. 1. Associations between wildfire-related $PM_{2.5}$ exposures and autism spectrum disorder (ASD). (A) Odds ratios (ORs) and 95% confidence intervals (CIs) for ASD associated with percentile-based wildfire $PM_{2.5}$ exposure during the entire pregnancy, shown separately for total wildfire $PM_{2.5}$, the biomass burning tracer, and the wildland–urban interface (WUI) tracer. Percentile groups (31st–60th, 61st–90th, and > 90th) are defined relative to the distribution of each exposure, with ≤ 30th percentile serving as the reference category (dashed line at OR = 1.0). (B) ORs and 95% CIs for ASD in relation to wildfire day counts and episode-based wildfire $PM_{2.5}$ exposures at concentrations ≥ 35 µg/m^3, evaluated for the entire pregnancy and the first year post-pregnancy. Episode-based metrics used 3-day and 7-day washout periods, requiring consecutive days below the threshold to define distinct wildfire smoke events. All models were adjusted for maternal, infant, and neighborhood-level covariates as described in the Methods. Corresponding numerical estimates are provided in Table S4.

61st–90th percentile, and > 90th percentile groups. Event-based metrics at the 35 µg/m^3 threshold also showed consistent associations across sexes. For males, the strongest effect estimate was seen for entire-pregnancy episodes with a 7-day washout (OR: 1.07; 95% CI: 1.06, 1.09), while for females the estimate was a slightly lower (OR: 1.04; 95% CI: 1.01, 1.08). No meaningful sex differences were observed for lower-intensity metrics or continuous WF-$PM_{2.5}$ and biomass burning exposures.

Associations between wildfire-related $PM_{2.5}$ and ASD were overall similar by socioeconomic status with a couple of deviations (Table S9). Among children from low SES areas, associations were generally null or inverse across most metrics, including continuous and percentile-based exposures. In contrast, in low-middle and middle SES groups, positive associations emerged at the highest exposure levels. For example, > 90th percentile entire-pregnancy wildfire exposure was associated with a 7–8% increase in odds of ASD. Associations were strongest in the high SES groups for WUI-related exposures and high-intensity events (≥ 35 µg/m^3).

Associations between wildfire-related $PM_{2.5}$ and ASD varied across racial/ethnic groups (Table S10). Among White children, consistent positive associations were observed, particularly for entire-pregnancy exposures at the highest percentiles and for high-intensity events (≥ 35 µg/m^3). For example, > 90th percentile entire-pregnancy wildfire exposure was associated with a 14% increase in odds of ASD (OR: 1.14; 95% CI: 1.10, 1.19). By contrast, among Hispanic children, associations were inverse across nearly all metrics. However, for WUI-related exposures, particularly the > 90th percentile and the ≥ 35 µg/m^3 episode-based metrics, elevated odds of ASD were observed across all racial/ethnic groups.

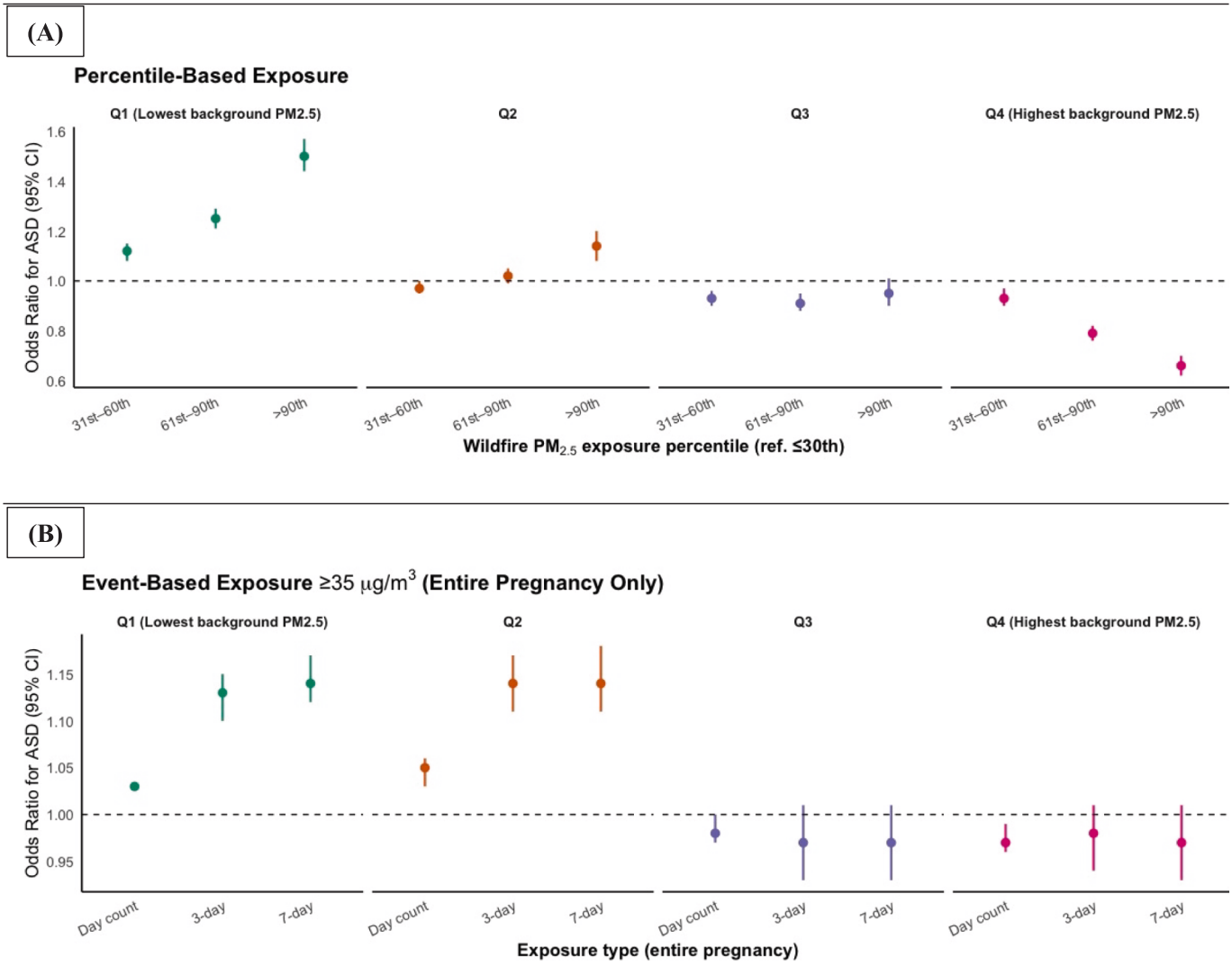


Fig. 2. Wildfire-Related $PM_{2.5}$ Exposure and ASD Risk Stratified by Background Air Pollution (A) Odds ratios (ORs) and 95% confidence intervals (CIs) for ASD in relation to percentile-based wildfire-related $PM_{2.5}$ exposure during pregnancy, comparing moderate (31st–60th), high (61st–90th), and extreme (> 90th) exposure categories. Estimates are stratified by quartiles of background (non-wildfire) $PM_{2.5}$ (Q1–Q4). (B) ORs and 95% CIs for event-based wildfire $PM_{2.5}$ exposures ≥ 35 μg/m^3 during pregnancy, based on cumulative wildfire day counts and episode-based metrics using 3-day and 7-day washout periods to define distinct wildfire smoke events. Results are stratified by background $PM_{2.5}$ quartiles. All models were adjusted for maternal, infant, and neighborhood-level covariates as described in the Methods. Corresponding numerical estimates are provided in Table S5.

4. Discussion

We observed increased ASD risk associated with wildfire smoke exposure during pregnancy, particularly for high-intensity events and concentrations in the uppermost exposure percentiles (> 90th percentile). These patterns suggest that wildfire-related exposures may pose a measurable neurodevelopmental risk at the highest concentrations of exposure or for certain chemical profiles, which may differ both spatially and temporally. This study highlights the importance of considering the complex and evolving nature of wildfire emissions and population vulnerability.

In overall models, wildfire smoke exposure was not associated with increased ASD risk when examined using continuous IQR-scaled metrics. Small positive associations emerged during the entire-pregnancy period when models were adjusted for non-wildfire $PM_{2.5}$ mass, consistent across both the total WF-$PM_{2.5}$ and the biomass burning tracer, reflecting the dominant contribution of biomass burning to total wildfire-related $PM_{2.5}$ mass. To our knowledge, there are no prior studies evaluating wildfire-specific $PM_{2.5}$ in relation to ASD risk, but these effect estimates appear smaller than those typically reported for general $PM_{2.5}$ mass exposure (Dutheil et al., 2021). As expected, overall $PM_{2.5}$ mass in this study was associated with a stronger increase in ASD risk, likely reflecting contributions from multiple urban emission sources previously implicated in ASD risk (O'Sharkey et al., 2025; O'Sharkey, Mitra, et al., 2024).

While wildfire-related biomass burning has not been examined directly in relation to ASD, prior research on biomass smoke provides a mechanistic foundation for our hypothesis. Biomass-derived particulate matter contains high levels of organic carbon, polycyclic aromatic compounds, and reactive species that can induce systemic inflammation and disrupt fetal brain development (Reid et al., 2016). Epidemiologic studies of biomass smoke exposure have similarly reported adverse

neurological outcomes (in adolescents), including disrupted cortical microarchitecture (Bottenhorn et al., 2024), while others reported no association with prenatal biomass combustion exposure (Luglio et al., 2024). Although our biomass burning tracer does not fully separate wildfire emissions from residential wood combustion, our wildfire exposure assessments improve substantially on prior work by leveraging a chemically resolved, source-informed chemical transport model integrated with ground observations. This hybrid CTM–Random Forest model incorporates satellite-derived burned area data and daily wildfire perimeters, allowing us to better characterize wildfire-related biomass combustion than traditional studies relying on undifferentiated $PM_{2.5}$ mass or general biomass smoke indicators.

In contrast to the continuous IQR-scaled metrics, percentile-based and episode-specific wildfire exposure measures revealed stronger and more consistent associations with ASD risk, particularly at the highest exposure levels. These approaches may better reflect the impact of rare but severe wildfire smoke events that are not captured in population-wide averages. Among the percentile-based exposure metrics, only the WUI tracer showed an overall association with ASD during entire-pregnancy exposure, suggesting that this component may reflect a more toxic mixture, including elevated levels of metals, synthetic organic compounds, and combustion by-products from building materials and infrastructure (National Academies and of Sciences et al., 2022). Wildfire smoke episodes exceeding 35 $\mu g/m^3$ were associated with increased ASD risk over the entirety of pregnancy, whereas the lower the threshold showed null or inverse associations. These findings suggest that intensity, rather than frequency alone, may be a more relevant driver of wildfire-related neurodevelopmental risk.

When stratified by background air pollution levels, we observed the strongest associations between wildfire smoke exposure and ASD risk among children in the lowest quartile of background $PM_{2.5}$. In this group, multiple wildfire exposure metrics, including continuous, percentile-based, and episode-specific measures, were consistently associated with elevated ASD odds. The consistency across metrics in this lowest exposure quartile suggests that wildfire smoke may be more impactful when overall pollutant burden is low and wildfire emissions represent a distinct acute source. However, given the very low absolute concentrations of the WUI tracer and the lack of association in continuous exposure models, caution is warranted when interpreting the modest elevation in risk observed across WUI exposure percentiles. These WUI-specific findings may reflect threshold effects, episodes involving more toxic combustion sources, or unmeasured spatial or temporal clustering of exposure. In contrast, among children exposed to high background pollution (Q3-Q4), wildfire metrics generally showed null or inverse associations with ASD. This may reflect reduced contrast between wildfire emissions and consistently elevated background pollution, survival bias for pregnancies experiencing multiple pollutants (Raz et al., 2018), or competing sources of air toxics, such as traffic and industrial emissions, which are major contributing sources of $PM_{2.5}$ and have been identified as risk factors for ASD (O'Sharkey et al., 2025).

These findings are further supported by our urbanicity-stratified results, which revealed elevated wildfire-ASD associations in non-metro areas, where background air pollution levels were lower on average. Although non-metro births represent a small share of the cohort (~5%), they tended to experience higher wildfire-related exposures at the upper end of the distribution, which may partly contribute to the stronger associations observed in these regions. Yet, even though non-metro regions are often characterized by lower background air pollution, this is not universally true as some rural areas, such as CA's Central Valley, experience persistently high pollution levels due to agricultural and traffic-related sources. In contrast, associations were consistently attenuated in metro areas where baseline $PM_{2.5}$ from urban sources such as traffic and industry is typically higher. These attenuated metro-area associations should not be interpreted as chronic urban pollution being protective; rather, several factors may reduce the observable impact of wildfire smoke in high-PM settings. For example, in high background air pollution settings, the cumulative burden of routine environmental insults plus wildfire smoke may contribute to increased pregnancy loss, potentially resulting in attenuated wildfire smoke effects among surviving live births. Live birth bias occurs when fetuses most susceptible to environmental exposures are lost (often early in pregnancy), causing the remaining population of live births to appear less affected. In areas with chronically elevated background pollution, this selection process may be even stronger when wildfire exposures are added. That is, wildfire smoke associations are underestimated because the most vulnerable pregnancies are no longer represented. These findings underscore the importance of considering the background pollutant context when evaluating the health impacts of episodic exposures like wildfire smoke.

Similar to the overall population, metropolitan areas showed a clear association between wildfire $PM_{2.5}$ from the WUI component and increased ASD risk, particularly at higher exposure percentiles. This pattern may reflect the unique chemical profile of smoke generated in developed zones, where high-temperature combustion of infrastructure, household materials, and synthetic compounds along with organic materials introduces additional neurotoxic constituents and releases toxic volatile organic compounds that can show serious health impacts (Dickinson et al., 2022). These exposures may exert effects independent of or in combination with urban background pollution and merit further toxicological investigation.

We observed differences in associations when stratifying the data according to the introduction of DSM-5 diagnostic criteria into the California DDS system in November 2014. While findings suggest stronger associations in the 2014–2019 birth cohorts, these differences may partly reflect changes in diagnostic practices following the adoption of DSM-5. They may also reflect broader shifts in ASD identification, including increased awareness, screening, and access to services that began prior to the formal DSM transition (O'Sharkey et al. (2024)). Importantly, these differences were not explained by higher wildfire exposures in the later period, since overall exposure distributions were similar across years. However, our data end in 2019 and therefore does not capture the extreme wildfire seasons that followed, including the 2020 August Complex Fire and 2021 Dixie Fire, two of the largest in California history, nor more recent urban-interface events such as the 2025 Palisades and Eaton Canyon fires. This underscores the importance of continued monitoring of birth cohorts exposed during these increasingly intense and geographically expansive wildfire seasons.

We also observed heterogeneity in associations by both socioeconomic status and race/ethnicity. Among children from low SES areas and Hispanic children, associations between wildfire-related $PM_{2.5}$ and ASD were generally null or inverse. In contrast, in White children and in middle- to higher-SES groups, positive associations emerged, particularly for entire-pregnancy exposures at the highest percentiles and for high-intensity events ($\geq$ 35 $\mu g/m^3$). These differences may partly reflect variation in diagnostic access, since children in lower SES and high pollution communities often face barriers to evaluation, which can lead to under-ascertainment compared to more affluent or White populations. However, associations with WUI-related exposures, particularly the $>$ 90th percentile and the $\geq$ 35 $\mu g/m^3$ episode-based metrics, remained elevated across all racial/ethnic and SES groups. This may suggest that extreme and spatially concentrated wildfire events occurring at the wildland–urban interface represent sufficiently severe exposures to drive measurable increases in ASD risk regardless of differential diagnostic access. Although higher income families may have greater capacity to temporarily relocate or use additional protective measures during severe wildfire smoke events (Burke et al., 2022), major wildfire episodes in California typically affect large regional air basins for extended periods, making ambient wildfire $PM_{2.5}$ a relevant exposure across socioeconomic groups even when individual avoidance behaviors differ.

Sex differences in susceptibility to air pollution and ASD risk have been reported in prior studies, with some suggesting that male children

may be more vulnerable to neurodevelopmental impacts of environmental exposures (Jo et al., 2019). In our analysis, wildfire-related $PM_{2.5}$ exposures were associated with similar ASD risk patterns across sexes, with slightly stronger associations observed among males for several wildfire exposure metrics, particularly in the > 90th percentile of WUI-related $PM_{2.5}$ and entire-pregnancy wildfire episodes $\geq 35\ \mu g/m^3$. These patterns suggest a possible heightened sensitivity among males, though the effect magnitude differences were modest and confidence intervals of effect estimates largely overlapped by sex. One possible reason for this is that male fetuses may be more sensitive to inflammation causing exposures such as $PM_{2.5}$ because male brains exhibit greater microglial abundance during prenatal and early postnatal development, a period when these immune cells are actively shaping neural circuits (Bolton et al., 2017; Hanamsagar & Bilbo, 2016).

This study has several notable strengths. First, the large, population-based cohort of more than 8.6 million births in California from 2001 to 2019 offers exceptional statistical power and supports generalizability within a diverse and wildfire-prone U.S. state. A major strength lies in the use of wildfire-specific $PM_{2.5}$ estimates derived from chemically resolved source tracers, which enabled us to distinguish wildfire contributions, including biomass burning and WUI smoke, from background $PM_{2.5}$. This multi-metric design allowed us to evaluate both average and acute exposure patterns. Furthermore, we conducted stratified analyses by background $PM_{2.5}$, urbanicity, ASD diagnostic era, and infant sex, allowing us to explore differential vulnerability and test hypotheses about modifying factors. Nonetheless, several limitations should be noted. First, exposure assignment was based on maternal residence at delivery, which may not capture residential mobility during pregnancy and could lead to exposure misclassification, particularly during wildfire events, when evacuations may displace those living closest to the fires. Next, biomass burning tracers likely capture not only wildfire emissions but also residential wood combustion, particularly during winter months, which we were unable to disentangle in the current analysis. We additionally do not have data on indoor air quality, time spent outdoors, and behaviors taken to minimize exposure, leading to further potential for exposure misclassification. It is important to note that our reliance on DDS data may skew the study population toward lower SES families who are more likely to access state services, potentially limiting generalizability and influencing observed associations. However, over-representation of lower-SES families in DDS does not preclude under-ascertainment within these communities; barriers to evaluation may still result in ASD cases not being diagnosed, which can attenuate community-specific associations if non-differential underdiagnosis is greater in low-SES communities. Additionally, high-intensity wildfire events may lead to temporary displacement, resulting in exposure misclassification if individuals were not present at their recorded birth address during smoke episodes. Lastly, while California provides a rich environment for studying wildfire smoke health effects, the findings may not fully generalize to other regions with different fire regimes, fuel types, or social determinants of health. Despite these limitations, this study provides a comprehensive and methodologically rigorous evaluation of wildfire-related $PM_{2.5}$ and ASD risk, and contributes novel insights into exposure timing, intensity, and source-specific toxicity in the context of neurodevelopment.

Overall, our findings suggest that wildfire smoke exposure during pregnancy is associated with increased ASD risk, particularly in contexts with lower background air pollution and among populations less chronically exposed to other sources of air toxics. However, it is also possible that in higher pollution settings, wildfire smoke may contribute to pregnancy loss, introducing live birth bias that could attenuate observed associations. Associations were strongest for high-intensity, episodic wildfire exposures and among children in non-metro areas, including those in WUI zones, highlighting the importance of both exposure intensity and underlying vulnerability. While some subgroup differences emerged, patterns were generally consistent, reinforcing the potential neurodevelopmental impact of wildfire-related air pollution. Additional studies minimizing exposure misclassification, including incorporating information on exposure-minimizing behaviors, clinically assessing autism outcomes, and investigating mechanisms would strengthen the robustness of these potentially public health-relevant findings.

CRediT authorship contribution statement

Karl O'Sharkey: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sanjali Mitra:** Writing – review & editing, Data curation. **Ting Chow:** Writing – review & editing, Data curation. **Amanda J. Goodrich:** Writing – review & editing, Methodology, Investigation, Data curation. **Kathryn C. Conlon:** Writing – review & editing, Data curation, Conceptualization. **Miriam Nuño:** Writing – review & editing, Conceptualization. **Deborah H. Bennett:** Writing – review & editing, Conceptualization. **Sean Raffuse:** Writing – review & editing, Methodology, Data curation. **Mohammad Astaneh:** Methodology, Data curation. **Yusheng Zhao:** Methodology, Data curation. **Michael Kleeman:** Writing – review & editing, Software, Methodology, Data curation. **Beate Ritz:** Writing – review & editing, Supervision, Project administration, Investigation, Data curation, Conceptualization. **Rebecca J. Schmidt:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Funding

EPA STAR 84048401; K99ES037743

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [RJS and DHB consult for Linus BioTechnology, Inc. RJS received research funding support from the Bia-Echo Foundation (12/1/2023 – 11/31/2025). RJS has received travel support to serve on the Observational Study Monitoring Board (OSMB) for HEALthy Brain and Child Development (HBCD) Study and to present at the Society for Birth Defects Research and Prevention 64th Annual Meeting (Jun 30, 2024), the Japan Society of Endocrine Disruptors Research in Tokyo, Japan (Dec 6, 2024), and the 2025 BCE Lung Foundation Air Quality and Health Workshop, Vancouver, British Columbia, Canada (Apr 8, 2025). The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper].

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envint.2026.110131.

Data availability

Data will be made available on request.

References

American Psychiatric Association. (2020). *Diagnostic and Statistical Manual of Mental Disorders*. DSM Library. https://dsm.psychiatryonline.org/doi/book/10.1176/appi.books.9780890425596.

Basilio, E., Chen, R., Fernandez, A.C., Padula, A.M., Robinson, J.F., Gaw, S.L., 2022. Wildfire smoke exposure during pregnancy: a review of potential mechanisms of placental toxicity, impact on obstetric outcomes, and strategies to reduce exposure. Int. J. Environ. Res. Public Health 19 (21), 13727. https://doi.org/10.3390/ijerph192113727.

Becerra, T.A., Wilhelm, M., Olsen, J., Cockburn, M., Ritz, B., 2013. Ambient air pollution and autism in los angeles county, California. Environ. Health Perspect. 121 (3), 380–386. https://doi.org/10.1289/ehp.1205827.

Beckerman, B., Jerrett, M., Brook, J.R., Verma, D.K., Arain, M.A., Finkelstein, M.M., 2008. Correlation of nitrogen dioxide with other traffic pollutants near a major expressway. Atmos. Environ. 42 (2), 275–290. https://doi.org/10.1016/j.atmosenv.2007.09.042.

Bolton, J.L., Marinero, S., Hassanzadeh, T., Natesan, D., Le, D., Belliveau, C., Mason, S. N., Auten, R.L., Bilbo, S.D., 2017. Gestational exposure to air pollution alters cortical volume, microglial morphology, and microglia-neuron interactions in a sex-specific manner. Front. Synaptic Neurosci. 9, 10. https://doi.org/10.3389/fnsyn.2017.00010.

Bottenhorn, K.L., Sukumaran, K., Cardenas-Iniguez, C., Habre, R., Schwartz, J., Chen, J.-C., Herting, M.M., 2024. Air pollution from biomass burning disrupts early adolescent cortical microarchitecture development. Environ. Int. 189, 108769. https://doi.org/10.1016/j.envint.2024.108769.

Brown, A.P., Cai, L., Laufer, B.I., Miller, L.A., LaSalle, J.M., Ji, H., 2022. Long-term effects of wildfire smoke exposure during early life on the nasal epigenome in rhesus macaques. Environ. Int. 158, 106993. https://doi.org/10.1016/j.envint.2021.106993.

Burke, M., Heft-Neal, S., Li, J., Driscoll, A., Baylis, P., Stigler, M., Weill, J.A., Burney, J. A., Wen, J., Childs, M.L., Gould, C.F., 2022. Exposures and behavioural responses to wildfire smoke. Nat. Hum. Behav. 6 (10), 1351–1361. https://doi.org/10.1038/s41562-022-01396-6.

Capitanio, J.P., Del Rosso, L.A., Gee, N., Lasley, B.L., 2022. Adverse biobehavioral effects in infants resulting from pregnant rhesus macaques' exposure to wildfire smoke. Nat. Commun. 13 (1), 1774. https://doi.org/10.1038/s41467-022-29436-9.

Cappa, C.D., Jathar, S.H., Kleeman, M.J., Docherty, K.S., Jimenez, J.L., Seinfeld, J.H., Wexler, A.S., 2016. Simulating secondary organic aerosol in a regional air quality model using the statistical oxidation model – Part 2: Assessing the influence of vapor wall losses. Atmos. Chem. Phys. 16 (5), 3041–3059. https://doi.org/10.5194/acp-16-3041-2016.

CARB. (n.d.). *Inhalable Particulate Matter and Health (PM2.5 and PM10) | California Air Resources Board*. Retrieved June 20, 2025, from https://ww2.arb.ca.gov/resources/inhalable-particulate-matter-and-health.

Carter, W.P.L., Heo, G., 2013. Development of revised SAPRC aromatics mechanisms. Atmos. Environ. 77, 404–414. https://doi.org/10.1016/j.atmosenv.2013.05.021.

Chiu, Y.-H.-M., Hsu, H.-H.-L., Coull, B.A., Bellinger, D.C., Kloog, I., Schwartz, J., Wright, R.O., Wright, R.J., 2016. Prenatal particulate air pollution and neurodevelopment in urban children: examining sensitive windows and sex-specific associations. Environ. Int. 87, 56–65. https://doi.org/10.1016/j.envint.2015.11.010.

Chun, H., Leung, C., Wen, S.W., McDonald, J., Shin, H.H., 2020. Maternal exposure to air pollution and risk of autism in children: a systematic review and meta-analysis. Environ. Pollut. 256, 113307. https://doi.org/10.1016/j.envpol.2019.113307.

Croen, L.A., Grether, J.K., Hoogstrate, J., Selvin, S., 2002. The changing prevalence of autism in California. J. Autism Dev. Disord. 32 (3), 207–215. https://doi.org/10.1023/a:1015453830880.

Cunningham, C.X., Williamson, G.J., Bowman, D.M.J.S., 2024. Increasing frequency and intensity of the most extreme wildfires on Earth. Nat. Ecol. Evol. 8 (8), 1420–1425. https://doi.org/10.1038/s41559-024-02452-2.

Dickinson, G.N., Miller, D.D., Bajracharya, A., Bruchard, W., Durbin, T.A., McGarry, J.K. P., Moser, E.P., Nuñez, L.A., Pukkila, E.J., Scott, P.S., Sutton, P.J., Johnston, N.A.C., 2022. Health risk implications of volatile organic compounds in wildfire smoke during the 2019 FIREX-AQ campaign and beyond. GeoHealth 6 (8), e2021GH000546. https://doi.org/10.1029/2021GH000546.

Dobois, E., & Sanders, A. (n.d.). *Rural-Urban Commuting Area Codes | Economic Research Service*. Retrieved June 20, 2025, from https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes.

Dutheil, F., Comptour, A., Morlon, R., Mermillod, M., Pereira, B., Baker, J.S., Charkhabi, M., Clinchamps, M., Bourdel, N., 2021. Autism spectrum disorder and air pollution: a systematic review and meta-analysis. Environ. Pollut. 278, 116856. https://doi.org/10.1016/j.envpol.2021.116856.

Fang, T., H. Hwang, B. C., Kapur, S., S. Hopstock, K., Wei, J., Nguyen, V., A. Nizkorodov, S., & Shiraiwa, M. (2023). Wildfire particulate matter as a source of environmentally persistent free radicals and reactive oxygen species. *Environmental Science: Atmospheres*, *3*(3), 581–594. 10.1039/D2EA00170E.

Hanamsagar, R., Bilbo, S.D., 2016. Sex differences in neurodevelopmental and neurodegenerative disorders: Focus on microglial function and neuroinflammation during development. The Journal of Steroid Biochemistry and Molecular Biology, SI: Steroids & Nervous System 160, 127–133. https://doi.org/10.1016/j.jsbmb.2015.09.039.

Heaney, A., Stowell, J.D., Liu, J.C., Basu, R., Marlier, M., Kinney, P., 2022. Impacts of fine particulate matter from wildfire smoke on respiratory and cardiovascular health in California. GeoHealth 6 (6), e2021GH000578. https://doi.org/10.1029/2021GH000578.

Hu, J., Jathar, S., Zhang, H., Ying, Q., Chen, S.-H., Cappa, C.D., Kleeman, M.J., 2017. Long-term particulate matter modeling for health effect studies in California – Part 2: concentrations and sources of ultrafine organic aerosols. Atmos. Chem. Phys. 17 (8), 5379–5391. https://doi.org/10.5194/acp-17-5379-2017.

Hu, J., Zhang, H., Chen, S.-H., Wiedinmyer, C., Vandenberghe, F., Ying, Q., Kleeman, M. J., 2014. Predicting primary PM2.5 and PM0.1 trace composition for epidemiological studies in California. Environ. Sci. Technol. 48 (9), 4971–4979. https://doi.org/10.1021/es404809j.

Jathar, S.H., Cappa, C.D., Wexler, A.S., Seinfeld, J.H., Kleeman, M.J., 2015. Multi-generational oxidation model to simulate secondary organic aerosol in a 3-D air quality model. Geosci. Model Dev. 8 (8), 2553–2567. https://doi.org/10.5194/gmd-8-2553-2015.

Jo, H., Eckel, S. P., Wang, X., Chen, J.-C., Cockburn, M., Martinez, M. P., Chow, T., Molshatzki, N., Lurmann, F. W., Funk, W. E., Xiang, A. H., & McConnell, R. (2019). Sex-specific associations of autism spectrum disorder with residential air pollution exposure in a large Southern California pregnancy cohort. *Environmental Pollution (Barking, Essex: 1987)*, *254*(Pt A), 113010. 10.1016/j.envpol.2019.113010.

Kleeman, M.J., Robert, M.A., Riddle, S.G., Fine, P.M., Hays, M.D., Schauer, J.J., Hannigan, M.P., 2008. Size distribution of trace organic species emitted from biomass combustion and meat charbroiling. Atmos. Environ. 42 (13), 3059–3075. https://doi.org/10.1016/j.atmosenv.2007.12.044.

Laurent, O., Hu, J., Li, L., Kleeman, M.J., Bartell, S.M., Cockburn, M., Escobedo, L., Wu, J., 2016. Low birth weight and air pollution in California: which sources and components drive the risk? Environ. Int. 92–93, 471–477. https://doi.org/10.1016/j.envint.2016.04.034.

Link Plus | Registry Plus | CDC. (2023, March 20). https://www.cdc.gov/cancer/npcr/tools/registryplus/lp.htm.

Luglio, D.G., Kleeman, M.J., Yu, X., Lin, J.C., Chow, T., Martinez, M.P., Chen, Z., Chen, J.-C., Eckel, S.P., Schwartz, J., Lurmann, F., McConnell, R., Xiang, A.H., Rahman, M.M., 2024. Prenatal exposure to source-specific fine particulate matter and autism spectrum disorder. Environ. Sci. Technol. 58 (42), 18566–18577. https://doi.org/10.1021/acs.est.4c05563.

National Academies of Sciences, E., Studies, D. on E. and L., Technology, B. on C. S. and, & Wildfires, C. on the C. of U. (2022). The Chemistry of Fires at the Wildland-Uban Interface. In *The Chemistry of Fires at the Wildland-Urban Interface*. National Academies Press (US). https://www.ncbi.nlm.nih.gov/books/NBK588632/.

Nevison, C., Blaxill, M., Zahorodny, W., 2018. California autism prevalence trends from 1931 to 2014 and comparison to national ASD Data from IDEA and ADDM. J. Autism Dev. Disord. 48 (12), 4103–4117. https://doi.org/10.1007/s10803-018-3670-2.

Oroumiyeh, F., Jerrett, M., Del Rosario, I., Lipsitt, J., Liu, J., Paulson, S.E., Ritz, B., Schauer, J.J., Shafer, M.M., Shen, J., Weichenthal, S., Banerjee, S., Zhu, Y., 2022. Elemental composition of fine and coarse particles across the greater Los Angeles area: spatial variation and contributing sources. Environ. Pollut. 292, 118356. https://doi.org/10.1016/j.envpol.2021.118356.

O'Sharkey, K., Meng, Q., Mitra, S., Paik, S.-A., Liu, J., Shen, J., Thompson, L.K., Chow, T., Su, J., Cockburn, M., Weichenthal, S., Paulson, S.E., Jerrett, M., Ritz, B., 2024. Associations between brake and tire wear-related PM2.5 metal components, particulate oxidative stress potential, and autism spectrum disorder in Southern California. Environ. Int. 185, 108573. https://doi.org/10.1016/j.envint.2024.108573.

O'Sharkey, K., Mitra, S., Chow, T., Paik, S.-A., Thompson, L., Su, J., Cockburn, M., Ritz, B., 2025. Prenatal exposure to criteria air pollution and traffic-related air toxics and risk of autism spectrum disorder: a population-based cohort study of California births (1990-2018). Environ. Int. 201, 109562. https://doi.org/10.1016/j.envint.2025.109562.

O'Sharkey, K., Mitra, S., Paik, S.-A., Chow, T., Cockburn, M., Ritz, B., 2024. Trends in the prevalence of autism spectrum disorder in California: disparities by sociodemographic factors and region between 1990-2018. J. Autism Dev. Disord. https://doi.org/10.1007/s10803-024-06371-w.

Ostro, B., Hu, J., Goldberg, D., Reynolds, P., Hertz, A., Bernstein, L., Kleeman, M.J., 2015. Associations of mortality with long-term exposures to fine and ultrafine particles, species and sources: results from the California Teachers Study Cohort. Environ. Health Perspect. 123 (6), 549–556. https://doi.org/10.1289/ehp.1408565.

Rahman, M.M., Shu, Y.-H., Chow, T., Lurmann, F.W., Yu, X., Martinez, M.P., Carter, S.A., Eckel, S.P., Chen, J.-C., Chen, Z., Levitt, P., Schwartz, J., McConnell, R., Xiang, A.H., 2022. Prenatal exposure to air pollution and autism spectrum disorder: sensitive windows of exposure and sex differences. Environ. Health Perspect. 130 (1), 17008. https://doi.org/10.1289/EHP9509.

Raz, R., Kioumourtzoglou, M.-A., Weisskopf, M.G., 2018. Live-birth bias and observed associations between air pollution and autism. Am. J. Epidemiol. 187 (11), 2292–2296. https://doi.org/10.1093/aje/kwy172.

Reid, C.E., Brauer, M., Johnston, F.H., Jerrett, M., Balmes, J.R., Elliott, C.T., 2016. Critical review of health impacts of wildfire smoke exposure. Environ. Health Perspect. 124 (9), 1334–1343. https://doi.org/10.1289/ehp.1409277.

Ritz, B., Liew, Z., Yan, Q., Cuia, X., Virk, J., Ketzel, M., Raaschou-Nielsen, O., 2018. Air pollution and autism in Denmark. Environmental Epidemiology (Philadelphia, Pa.) 2 (4), e028.

Schlosser, J.S., Braun, R.A., Bradley, T., Dadashazar, H., MacDonald, A.B., Aldhaif, A.A., Aghdam, M.A., Mardi, A.H., Xian, P., Sorooshian, A., 2017. Analysis of aerosol composition data for western United States wildfires between 2005 and 2015: dust emissions, chloride depletion, and most enhanced aerosol constituents. J. Geophys. Res. Atmos. 122 (16), 8951–8966. https://doi.org/10.1002/2017JD026547.

Schollaert, C., Connolly, R., Cushing, L., Jerrett, M., Liu, T., Marlier, M., 2025. Air quality impacts of the January 2025 los angeles wildfires: insights from public data sources. Environ. Sci. Technol. Lett. 12 (8), 911–917. https://doi.org/10.1021/acs.estlett.5c00486.

Seinfeld, J. H., & pandis, S. (n.d.). *Atmospheric Chemistry And Physics* (2nd ed.). Wiley. Retrieved July 31, 2025, from http://archive.org/details/0237-pdf-atmospheric-chemistry-and-physics-2nd-ed-j.-seinfeld-s.-pandis-wiley-2006-ww.

Shaw, K. A., Williams, S., Patrick, M. E., Valencia-Prado, M., Durkin, M. S., Howerton, E. M., Ladd-Acosta, C. M., Pas, E. T., Bakian, A. V., Bartholomew, P., Nieves-Muñoz, N., Sidwell, K., Alford, A., Bilder, D. A., DiRienzo, M., Fitzgerald, R. T., Furnier, S. M., Hudson, A. E., Pokoski, O. M., … Maenner, M. J. (2025). Prevalence and Early Identification of Autism Spectrum Disorder Among Children Aged 4 and 8 Years—Autism and Developmental Disabilities Monitoring Network, 16 Sites, United

States, 2022. *MMWR Surveillance Summaries*, *74*(2), 1–22. 10.15585/mmwr.ss7402a1.

Urbanski, S. P., Hao, W. M., & Baker, S. (2008). Chapter 4 Chemical Composition of Wildland Fire Emissions. In A. Bytnerowicz, M. J. Arbaugh, A. R. Riebau, & C. Andersen (Eds.), *Developments in Environmental Science* (Vol. 8, pp. 79–107). Elsevier. 10.1016/S1474-8177(08)00004-1.

Venecek, M.A., Yu, X., Kleeman, M.J., 2019. Predicted ultrafine particulate matter source contribution across the continental United States during summertime air pollution events. Atmos. Chem. Phys. 19 (14), 9399–9412. https://doi.org/10.5194/acp-19-9399-2019.

Wegesser, T.C., Pinkerton, K.E., Last, J.A., 2009. California wildfires of 2008: coarse and fine particulate matter toxicity. Environ. Health Perspect. 117 (6), 893–897. https://doi.org/10.1289/ehp.0800166.

White, A.R., 2024. The firestorm within: a narrative review of extreme heat and wildfire smoke effects on brain health. Sci. Total Environ. 922, 171239. https://doi.org/10.1016/j.scitotenv.2024.171239.

Wilgus, M.-L., Merchant, M., 2024. Clearing the air: understanding the impact of wildfire smoke on asthma and COPD. Healthcare (Basel, Switzerland) 12 (3), 307. https://doi.org/10.3390/healthcare12030307.

Williams, A.P., Abatzoglou, J.T., Gershunov, A., Guzman-Morales, J., Bishop, D.A., Balch, J.K., Lettenmaier, D.P., 2019. Observed impacts of anthropogenic climate change on wildfire in California. Earth's Future 7 (8), 892–910. https://doi.org/10.1029/2019EF001210.

Willson, B.E., Gee, N.A., Willits, N.H., Li, L., Zhang, Q., Pinkerton, K.E., Lasley, B.L., 2021. Effects of the 2018 Camp Fire on birth outcomes in non-human primates: case-control study. Reprod. Toxicol. 105, 128–135. https://doi.org/10.1016/j.reprotox.2021.08.005.

Yang, L.-S., Kleeman, M.J., Cushing, L.J., Lipsitt, J., Su, J., Burnett, R.T., Batteate, C.M., Nau, C.L., Young, D.R., Tartof, S.Y., Butler, R.K., Padilla, A., Jerrett, M., 2024. Risks of source and species-specific air pollution for COVID-19 incidence and mortality in Los Angeles. Environ. Res.: Health 2 (4), 041001. https://doi.org/10.1088/2752-5309/ad67fb.

Yang, W., Hamilton, J.L., Kopil, C., Beck, J.C., Tanner, C.M., Albin, R.L., Ray Dorsey, E., Dahodwala, N., Cintina, I., Hogan, P., Thompson, T., 2020. Current and projected future economic burden of Parkinson's disease in the U.S. Npj Parkinson's Disease 6 (1). https://doi.org/10.1038/s41531-020-0117-1.

Yao, J., Brauer, M., Henderson, S.B., 2013. Evaluation of a wildfire smoke forecasting system as a tool for public health protection. Environ. Health Perspect. 121 (10), 1142–1147. https://doi.org/10.1289/ehp.1306768.

Yu, X., Venecek, M., Kumar, A., Hu, J., Tanrikulu, S., Soon, S.-T., Tran, C., Fairley, D., Kleeman, M.J., 2019. Regional sources of airborne ultrafine particle number and mass concentrations in California. Atmos. Chem. Phys. 19 (23), 14677–14702. https://doi.org/10.5194/acp-19-14677-2019.